
Freight Rate Prediction

A reproducible forecasting solution with explicit data-quality rules, two-month forward validation, and exact submission checks.

The solution fits on 48,000 labeled loads from January-October 2025 and produces one prediction for each of the 12,000 November-December loads. A separate core ensemble handles the fixed December scenario using the fields actually supplied.

Validation evidence

Rich ensemble MAE	Rich ensemble RMSE	Median absolute error
\$126.07	\$634.59	\$50.35

Scores pool 28,890 future-month predictions across three expanding-window folds. Dollar MAE is the primary selection criterion; RMSE remains visible to show tail risk. The folds informed model choice, so these are internal selection-validation results, not an untouched test score.

Model	MAE (\$)	RMSE (\$)
Training median	1,151.03	1,573.03
CatBoost with quote signal	133.54	636.95
Seasonal rich CatBoost	130.26	636.43
Robust rich regression	127.77	634.35
Selected rich ensemble	126.07	634.59
Selected core ensemble	137.23	636.11

Model and delivery

Selected rich model: equal-weight seasonal CatBoost and robust dollar regression. CatBoost captures nonlinear interactions; the robust component supplies smooth distance and seasonal effects. The combined MAE improves on either component while preserving a simple inference path.

Selected core model: equal-weight CatBoost on rate per mile and robust dollar regression. Distance weighting aligns both tree training and early stopping with dollar MAE. Both components are independently evaluated with the December-compatible feature contract.

Deliverables: reproducible Python code, one executed analysis notebook, pinned dependencies, the exact 12,000-row prediction CSV, a 31-row December CSV, validation evidence, and this report with the supplied scorer chart.

Data quality and feature logic

Preserve information, fit learned preprocessing inside each training partition, and respect the inference schema.

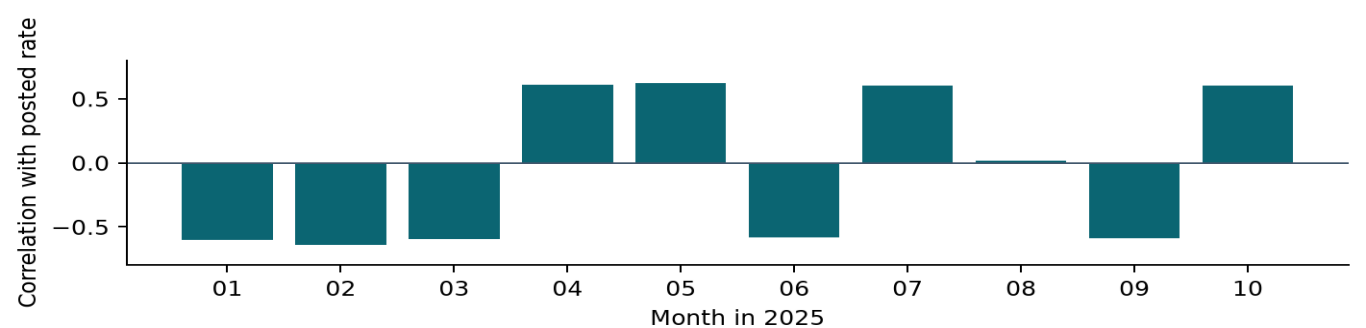
Check	Development	Final validation	Treatment
Missing weight	300	165	Retain NaN; train-fitted imputation in robust model.
Negative weight	292	145	Keep absolute magnitude and anomaly flag.
Missing market index	374	249	Native NaN handling in tree; train-fitted median in robust model.
Duplicate IDs / full rows	0 / 0	0 / 0	Check before fitting and export.
Rows with an unseen city	0	1,447	Population city fallback plus coordinates in rich tree.
Extreme posted rates	\$25,533 maximum	Targets unavailable	Retain every target; use robust losses.

Absolute negative-weight magnitudes resemble valid weights (mean approximately 31,724 lb versus 31,415 lb). This supports a sign-error assumption; it does not prove the source semantics. Original CSVs remain unchanged. No target outliers are trimmed.

Two explicit feature contracts

Feature group	Rich validation model	December core model
Pickup, delivery, lane, distance, equipment, weight	Available	Available
Weekday, day of month, annual sine/cosine, distance transforms	Derived from supplied fields	Derived from supplied fields
Pickup/delivery coordinates, market index	Available	Excluded; not supplied
Quote signal, load ID, target as a predictor	Excluded from selected model	Excluded

Quote-signal stability check



The quote relationship changes sign across months and weakens in August. A quote-inclusive tree is explicitly compared, but the selected ensemble avoids relying on a relationship whose future regime cannot be established from ten labeled months.

Leakage controls: no random split; no ID feature; no final-validation target; no global learned imputation, scaling, or city statistics. Early stopping and categorical statistics use only the training window. Future outcomes never enter any fit.

Validation and diagnostics

Every outer test window covers two complete future months, matching the November-December forecast horizon.

Training period	Train rows	Outer test period	Test rows
2025-01-01 to 2025-04-30	19,110	2025-05-01 to 2025-06-30	9,696
2025-01-01 to 2025-06-30	28,806	2025-07-01 to 2025-08-31	9,671
2025-01-01 to 2025-08-31	38,477	2025-09-01 to 2025-10-31	9,523

Within each training window, the last month is reserved for early stopping with 60 rounds of patience and a maximum of 600 trees. That tree count is then refitted on all rows in the window. The outer two-month outcomes are used for model selection and reporting, never for early stopping within that fold.

Outer fold	Rich ensemble MAE (\$)	Core ensemble MAE (\$)
May-June	134.55	138.71
July-August	104.36	134.69
September-October	139.46	138.31

Final refit: October supplies the final early-stopping window; then both ensembles refit on all 48,000 loads. Rich tree: 476 trees, depth 6. Core tree: 510 trees, depth 4. Both use learning rate 0.05, seed 42, and MAE loss. Huber regression uses epsilon 1.35 and L2 alpha 0.01. The blend is fixed at 50/50.

Errors by equipment

Equipment	Rows	MAE (\$)	RMSE (\$)
Dry Van	16,381	117.03	647.41
Flatbed	5,257	137.03	574.55
Reefer	7,252	138.52	646.70

The rich ensemble has a 90th-percentile absolute error of \$176.93. The largest 1% of absolute errors contribute 96.6% of total squared error. The gap between MAE and RMSE therefore reflects a small, severe tail. All rows remain in reported metrics.

Unseen-city stress test

Eight cities are withheld from January-June training, then later July-August loads involving those cities are scored. This leaves 25,726 training rows and 1,009 stress-test rows. Rich MAE: \$91.98; core MAE: \$102.58. This is an additional development stress test, not an independent final score.

Unknown city labels receive the training-population contribution in the robust component. The rich tree also has supplied geographic coordinates. This avoids dropping loads or silently treating an unseen city as a known reference city.

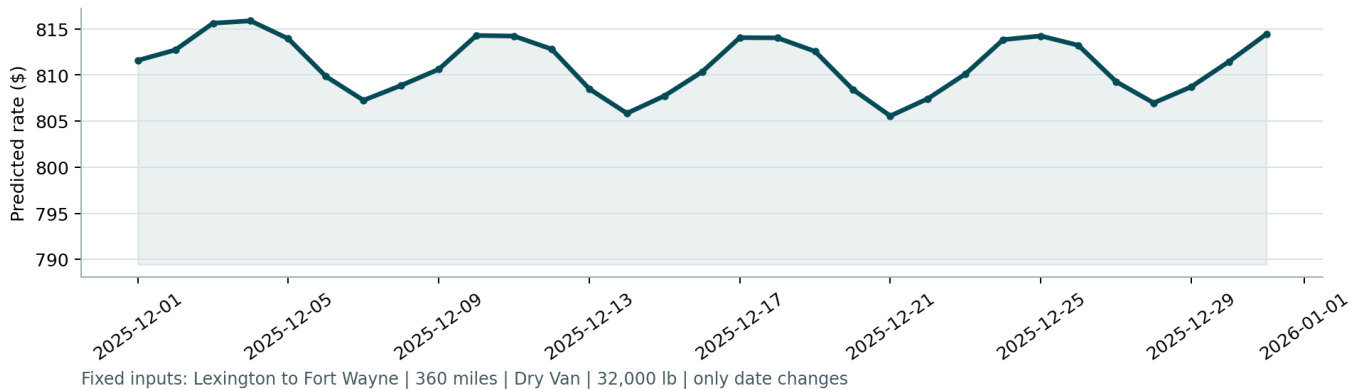
Fixed December prediction

Required chart generated with the supplied score.py plotting function.

Fixed lane: Lexington to Fort Wayne | **Distance:** 360 miles

Equipment: Dry Van | **Weight:** 32,000 lb | **Dates:** December 1-31, 2025

Candidate: December 2025 Predicted Load Rate



The core ensemble produces 31 finite, positive predictions, ranging from \$805.56 to \$815.87. Only the date varies. Coordinates and market fields are excluded throughout core training and inference. The curve is a model scenario, not a measured-rate comparison or a confidence band.

Reproducibility and output integrity

Run the following commands from the repository root after installing the pinned requirements. The first command reproduces the data audit, rolling validation, geographic stress test, final predictions, and chart. The second independently applies the supplied scorer.

```
python fit_submission.py
python score.py --predictions validation_predictions.csv --december-predictions december_predictions.csv
--output-dir results
```

The validation CSV has exactly two columns in the required order and exactly 12,000 unique template-aligned IDs. The December CSV preserves the seven original columns, the fixed scenario inputs, and all 31 dates. Input SHA-256 hashes, environment versions, split details, model parameters, and diagnostic metrics are recorded in results/validation_summary.json.

Practical limits

Only January-October of one year are labeled. Winter behavior and annual seasonality cannot be independently verified. Unseen locations, future market shifts, and rare extreme rates remain material sources of uncertainty. The supplied scorer validates structure and creates the chart; it has no hidden targets and returns no final accuracy metric.

Run instructions and model documentation are included in README.md. The executed notebook reproduces the same validation procedure and submission outputs using the shared implementation.

Implementation references: [CatBoost missing-value handling](#); [scikit-learn HuberRegressor](#).